CLASSIFICATION

Nearest-Neighbor classifiers

Eager Learners

- Designed to learn a model that maps the input attributes to the class label as soon as the training data becomes available
- Examples:decision tree classifiers, rule based classifiers

Lazy learners

- Delay the process of modelling the training data until it is needed to classify the test examples
- Example: Rote classifier - memorizes the entire training data and performs classification only if the attributes of a test instance match one of the training examples exactly
- Drawback:some test records may not be classified because they do not match any training example

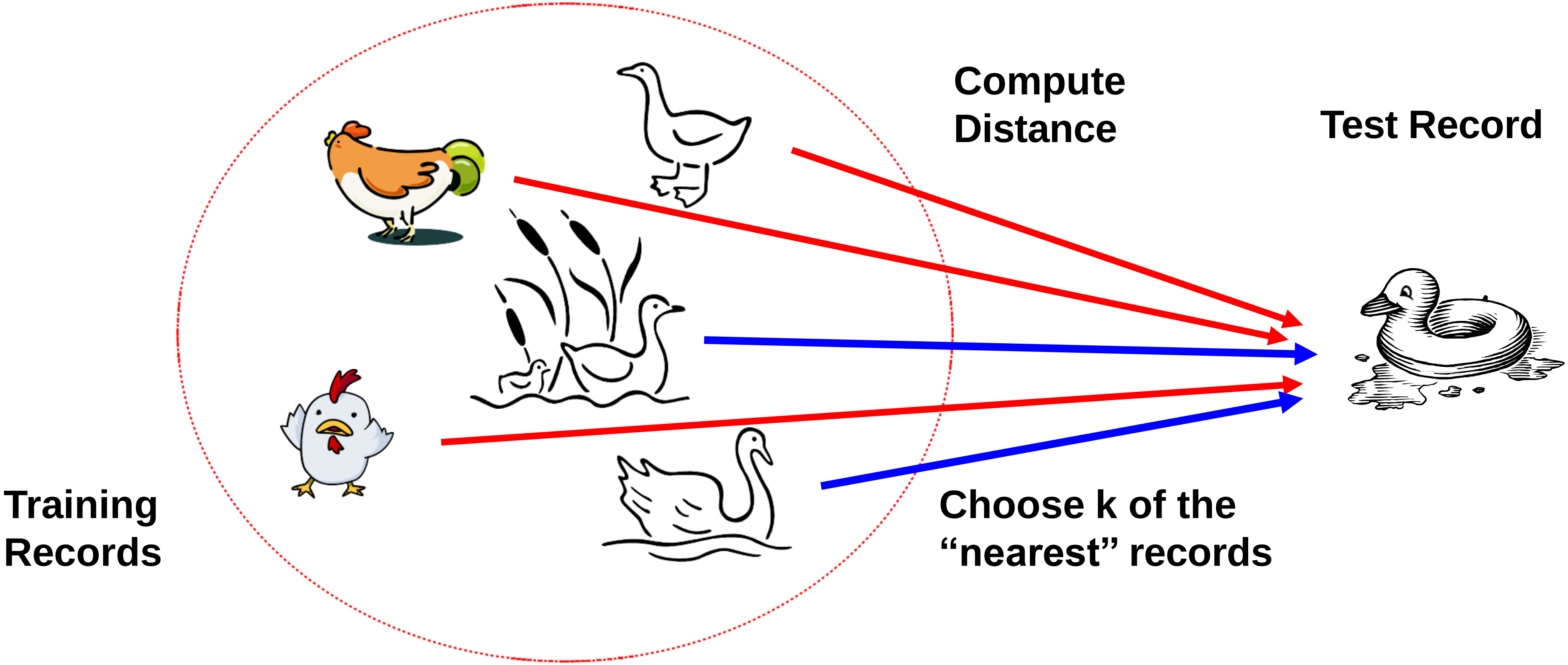
Solution

- Find all the training examples that are relatively similar to the attributes of the test example

- These nearest neighbours, can be used to determine the class label of the test example

Basic idea:

- If it walks like a duck, quacks like a duck, if it looks like duck, then it's probably a duck



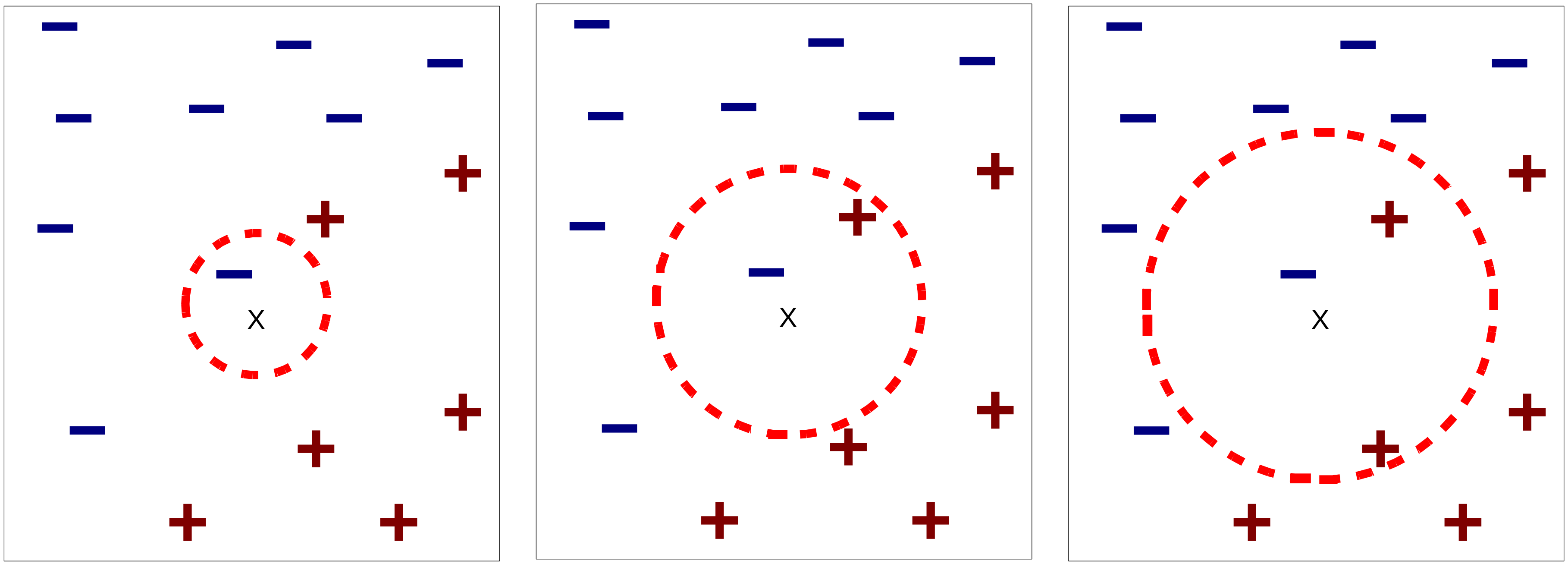
- A nearest neighbor classifier represents each example as a data point in a d-dimensional space, where d is the number of attributes

- Given a test example, we compute its proximity to the rest of the data points in the training set, using one of the proximity measure

- K-nearest neighbors of a given example z refer to the k points that are closest to z

- Data point is classified based on the class labels of its neighbors

- In the case where the neighbors have more than one label, the data point is assigned to the majority class of its nearest neighbors



(a) 1-nearest neighbor (b) 2-nearest neighbor (c) 3-nearest neighbor

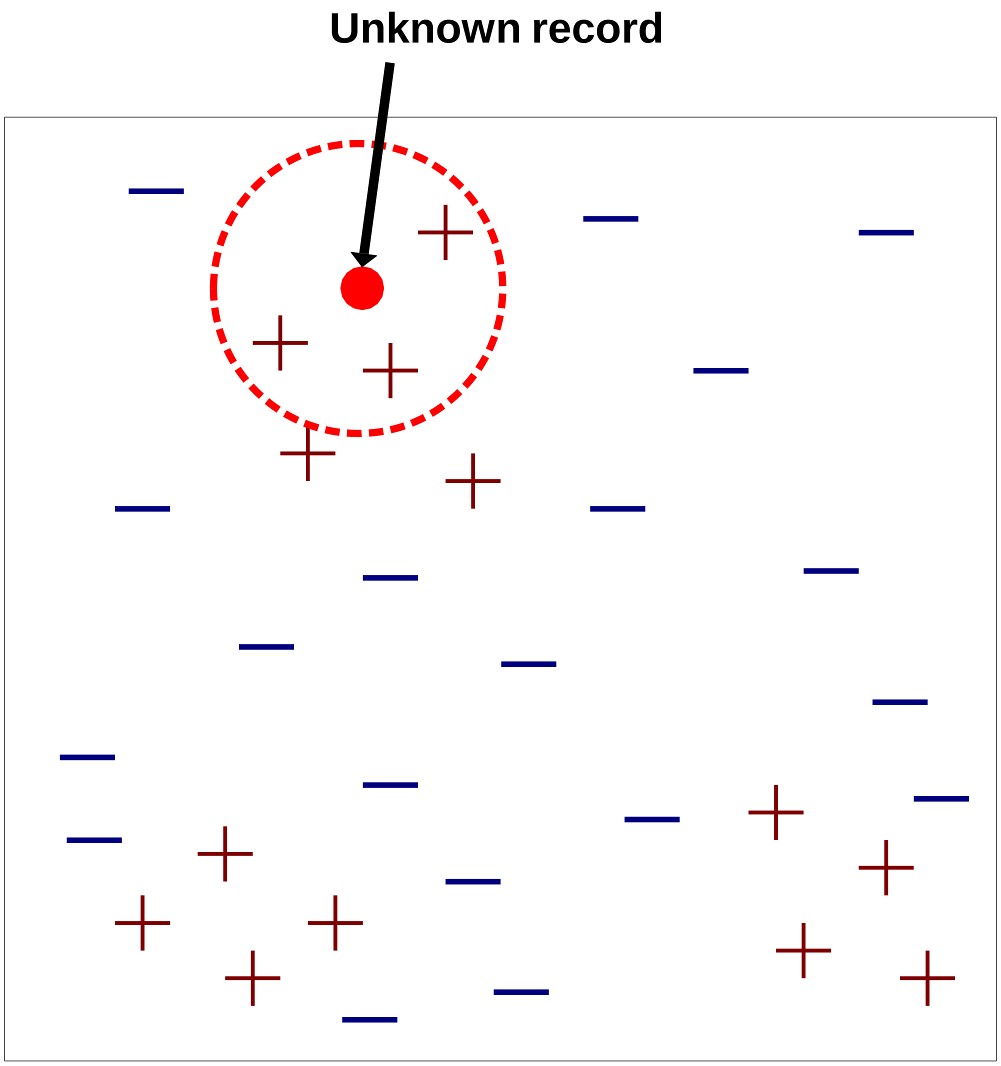
K-nearest neighbors of a record x are data points that have the k smallest distances to x

- If the number of nearest neighbors is one, data point is a negative example. Therefore the data point is assigned to the negative class

- If the number of nearest neighbors is three, then the neighborhood contains two positive examples and one negative example

- Using the majority voting scheme, the data point is assigned to the positive class

- In the case where there is a tie between the classes, we may randomly choose one of them to classify the data point Choose the right value for k



- Requires three things
  - The set of labeled records
  - Distance Metric to compute distance between records
  - The value of *k*, the number of nearest neighbors to retrieve

- To classify an unknown record:
  - Compute distance to other training records
  - Identify *k* nearest neighbors
  - Use class labels of nearest neighbors to determine the class label of unknown record (e.g., by taking majority vote)

Algorithm

- Algorithm computes the distance (or similarity) between each test example $z = (\mathbf{x}', y')$ training examples (x, y) belongs D to determine its nearest-neighbor list, D_z

- Such computation can be costly if the number of training examples is large

Let k be the number of nearest neighbors and D be the set of training examples.

for each test example $z = (\mathbf{x}', y')$ **do**

  Compute $d(\mathbf{x}', \mathbf{x})$, the distance between z and every example, $(\mathbf{x}, y) \in D$.

  Select $D_z \subseteq D$, the set of k closest training examples to z.

  $y' = \underset{v}{\operatorname{argmax}} \sum_{(\mathbf{x}_i, y_i) \in D_z} I(v = y_i)$

end for

- Once the nearest-neighbor list is obtained, the test example is classified based on the majority class of its nearest neighbors

$$\text{Majority Voting: } y' = \underset{v}{\operatorname{argmax}} \sum_{(\mathbf{x}_i, y_i) \in D_z} I(v = y_i),$$

- Where:
- v is a class label
- y_i is the class label for one of the nearest neighbors
- I is an indicator function that returns the value 1 if its argument is true and 0 otherwise

- In the majority voting approach, every neighbor has the same impact on the classification which makes the algorithm sensitive to the choice of k
- To reduce impact of k, weight the influence of each nearest x_i according to its distance: $w = 1/d^2$
- As a result, training examples that are located far away from *z* have a weaker impact the classification compared to those that are located close to *z*
- Using the distance weight voting scheme, the class label can be determined as:

$$y' = \underset{v}{\operatorname{argmax}} \sum_{(\mathbf{x}_i, y_i) \in D_z} w_i \times I(v = y_i)$$

Characteristics of Nearest Neighbor Classifier

- NN classification is part of instance based learning, which uses specific training instances to make predictions without having to maintain model derived from data
- Do not require model building
- Predictions are based on local information
- Can produce wrong predictions unless the appropriate proximity measure and data preprocessing steps are taken
- Can produce arbitrary shaped decision boundaries